

# QUANTUM PORTFOLIO CO-PILOT

WISER x Vanguard  
Quantum for Finance  
Challenge 2026

## Quantum-Enhanced Multi- Asset Portfolio Construction

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Audited classical optimization + exact QUBO/QAOA benchmark  
+ higher-moment HUBO/VQE research extension

ZERO HARD-CONSTRAINT BREACHES

EXACT CLASSICAL VALIDATION

QUANTUM-COMPATIBLE

[github.com/anurag-dot-physx/wiser-challenge-final](https://github.com/anurag-dot-physx/wiser-challenge-final)

*Claim boundary: quantum results are benchmarked against exact enumerable references; no quantum-advantage claim.*

# Challenge and scoring priorities

Optimize expected utility while preserving hard investment guardrails and interpretability.

## Portfolio utility

Expected return + variance risk + income + implementation cost + scenario/drawdown control.

## Hard guardrails

Fully invested, long-only, single-asset cap, equity maximum, defensive minimum, alternatives maximum.

## Quantum compatibility

Binary variables, linear constraints and a quadratic cost function that can be audited classically.

## What the solution must show

- Classical baseline, then constraints and scenario penalties
- Tunable growth, income, drawdown and cost sensitivity
- Risk/return/turnover/breach comparison with rationale
- Working co-pilot and classical validation

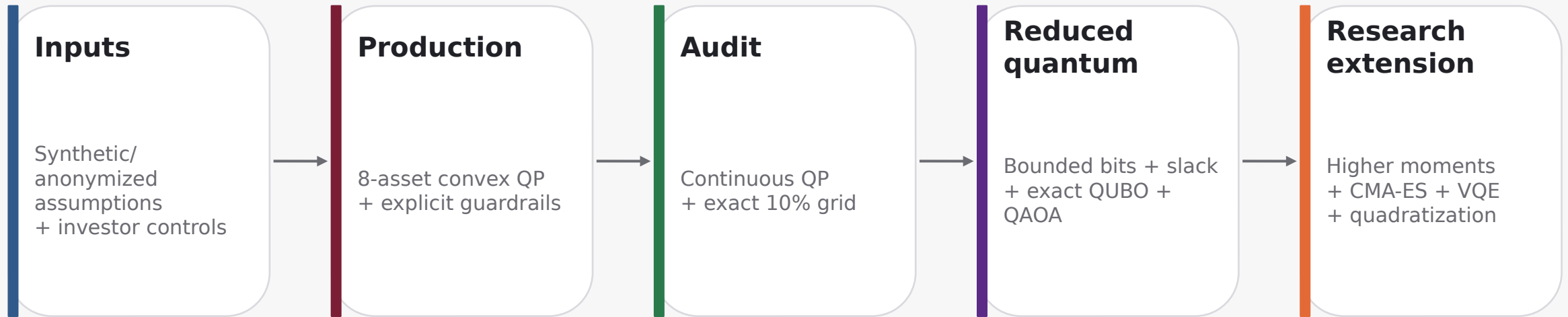
## Design principle

Feasibility first. Every production recommendation is post-validated; every quantum-compatible encoding is compared with an exact classical reference.

*Source: WISER <> Vanguard Quantum for Finance challenge brief.*

# Solution architecture

Separate the production recommendation from research benchmarks so comparisons remain valid.



## Key separation

The eight-asset production solution is never compared directly with the reduced 5-sleeve quantum encoding. Each quantum result is judged against its own exact optimum.

# Flagship model: convex quadratic utility + linear guardrails

$$J(w) = -\lambda \mu^T w + \lambda r w^T \Sigma w - \lambda y^T w + \lambda T (w - w_0)^T R (w - w_0) + \lambda s \cdot \text{mean}[(\ell s^T w)^2]$$

## Convexity audit

$\Sigma$ ,  $R$  and the scenario Gram matrix are positive semidefinite; the Hessian is positive semidefinite for every canonical profile.

## Hard linear guardrails

- $\sum_i w_i = 1$  and  $w_i \geq 0$
- single-asset maximum
- equity exposure maximum
- defensive exposure minimum
- alternatives exposure maximum

## Investor controls

Growth · risk · income ·  
drawdown · rebalancing  
sensitivity

## Reported metrics

Expected return · volatility · risk-  
adjusted ratio · turnover · cost ·  
scenario loss · breaches

# Classical audit: exact enumeration verifies the production model

The 10% grid contains 19,448 fully invested portfolios.

| Profile   | Form error | min eig(H) | QP→grid gap | QUBO vars | Feasible |
|-----------|------------|------------|-------------|-----------|----------|
| Growth    | 5.55e-17   | 1.010e-02  | 0.00081799  | 13        | 108      |
| Balanced  | 6.94e-17   | 2.185e-02  | 0.00066947  | 15        | 66       |
| Defensive | 4.86e-17   | 3.365e-02  | 0.00140581  | 11        | 2        |

**PASS**

Zero canonical hard breaches. QUBO ground portfolio matches its exact reduced financial optimum.

## Cost-sensitivity check

One-way turnover changes from 40% at  $\lambda T = 0$  to 15% at  $\lambda T = 2$ , demonstrating a material implementation-cost control.

## Interpretation

Objective decomposition matches matrix form to numerical precision; the continuous solution is a lower bound to the exact-grid objective, as expected.

# Reduced exact-constraint QUBO and QAOA benchmark

$$E(x,s) = J(Ax) + P \sum_k (c_k^T x + d_k^T s_k - t_k)^2$$

## Growth

13 vars  
108 feasible

## Balanced

15 vars  
66 feasible

## Defensive

11 vars  
2 feasible

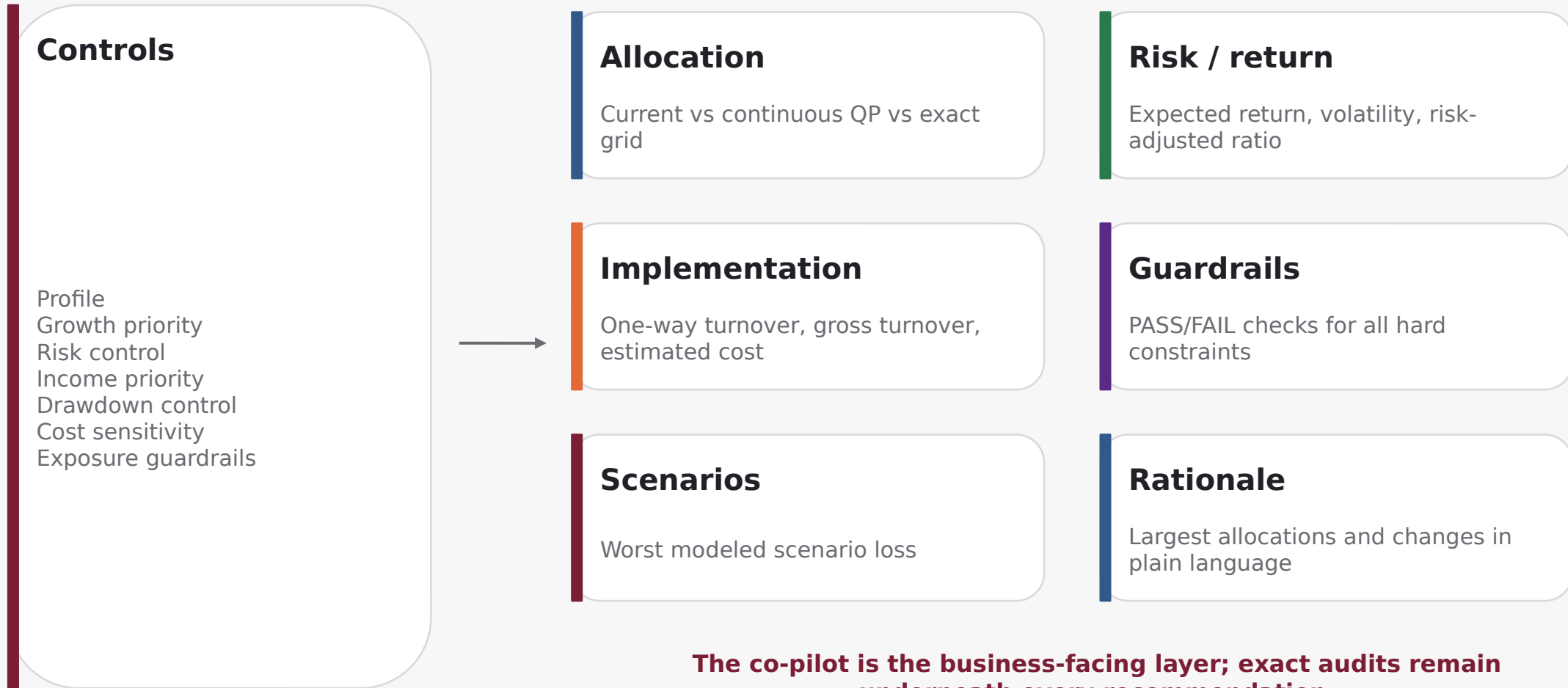
## Exhaustive checks

- constraint-consistent states imply hard-feasible portfolios
- unique consistent portfolios = unique hard-feasible portfolios
- global QUBO ground is hard-feasible
- global QUBO ground matches reduced exact optimum

## QAOA benchmark

Run only after the exact QUBO audit passes. Finite-shot post-processing selects among observed hard-feasible portfolios; fallback, probability mass and optimality gap are explicit.

# Portfolio co-pilot: optimization translated into investor-facing decisions



# Higher-moment extension: learn the Hamiltonian, then benchmark VQE

$$E(m) = -\lambda \mu \mu^T m + \lambda \Sigma m^T \Sigma m - \lambda S \Sigma S_{ijk} m_i m_j m_k + \lambda K \Sigma K_{ijkl} m_i m_j m_k m_l + \lambda B (\Sigma m_i - 8)^2$$

## 12 qubits

4 assets × 3 bits. Full source tensors. USD 1,250 unit makes 8 units exactly USD 10,000.

## 15 qubits

5 assets × 3 bits. Scaling demo; mixed TSLA+NVDA higher-moment terms are zero-imputed.

## TRAIN

CMA-ES learns  $\lambda$

## SELECT

Choose  $G^*$  on 5 validation sectors

## HOLD OUT

Freeze model; test 5 untouched sectors

## QUANTUM

Sweep final VQE budget

**Efficiency:** state features are cached once; VQE is intentionally outside the CMA-ES coefficient-training loop.



# Exact HUBO → QUBO quadratization

## Original HUBO

Cubic + quartic pseudo-Boolean interactions  
12 or 15 native allocation bits



## Rosenberg product constraint

Introduce  $y = ab$   
Add  $P(ab - 2ay - 2by + 3y)$   
Choose  $P$  above a rigorous objective range bound

## Energy equivalence

Every original state has identical energy after consistent ancilla lifting.

## Ground-state equivalence

Lifted QUBO ground portfolio is checked against the exact HUBO ground.

## Scalability trade-off

Higher-order couplers are removed at the cost of additional binary variables.

# Findings, limitations and reviewer path

## Established

Convex flagship objective  
Complete 19,448-state grid audit  
Zero canonical hard breaches  
Exact reduced-QUBO ground match

## Exploratory

QAOA/VQE solver quality  
Higher-moment calibration stability  
15-qubit stitched scaling  
Ancilla overhead

## Claim boundary

No production quantum speedup claim  
No economic outperformance claim  
No restricted Vanguard data  
Not investment advice

## Reproduce the flagship audit

```
git clone https://github.com/anurag-dot-physx/wiser-challenge-final.git
pip install -r requirements.txt && pip install -e .
pytest -q && python src/audit_quadratic_model.py
streamlit run src/vanguard_copilot_app.py
```

## Submission takeaway

Feasible first.  
Exact reference second.  
Quantum benchmark third.